



A hybrid approach for outlier detection in pharmaceutical cold chain logistics: A case study

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ABSTRACT

Pharmaceutical products are highly sensitive to environmental factors including temperature fluctuations, humidity, light and vibrations. Therefore, pharmaceutical cold chain disruptions pose significant human health, financial and regulatory risks, with logistics activities being the most critical point of vulnerability. This study proposes a hybrid approach called Route Detection-based Support Vector Regression (RD-SVR) algorithm for detecting temperature outliers during transportation of pharmaceutical products. The algorithm first conducts a preprocessing stage on the big data set where it detects the routes and prunes the unnecessary samples at the same time. Then, the cleaned data is used to train the SVR model for outlier detection. The automated route-based structure offers a real-time solution for cases involving a dynamic set of varying routes and vehicles. The proposed RD-SVR model is applied on one-year data logs of two vehicles gathered from an international logistics company, and tested against Random Forest and ANN. Test results and sensitivity analysis highlight the robustness and effectiveness of this innovative classification-based outlier detection model for identifying cold chain breakages and safeguarding product quality.

1. Introduction

A cold chain is a temperature-controlled supply chain that involves the production, storage and transportation of sensitive and perishable products such as fruits and vegetables, meat, seafood, dairy products, pharmaceuticals, and some chemicals. Cold chain storage and transportation involves many irreversible risks including product degradation and regulatory violations (Ding et al., 2023; Lin et al., 2023). Therefore, risks in cold chain storage and transportation must be thoroughly analyzed and proactive interventions should be implemented to prevent irreversible consequences. An effective cold chain management necessitates specialized packaging systems and vehicles, and continuous real time monitoring of environmental factors such as temperature, humidity, light, and vibration. The emergence of Industry 4.0 technologies and increasing digitalization of supply chains help to increase traceability of supply chain operations and develop data-driven approaches in cold chains.

Pharmaceutical cold chains, being a critical stakeholder of the cold chain, play a critical role for the healthcare industry for ensuring the

safety and efficacy of medications. Pharmaceutical products are highly sensitive to environmental factors such as extreme or fluctuating temperatures and humidity. Improper environmental conditions may lead to chemical degradation, compromise the efficacy of medications, and even cause the death of humans. Therefore, all phases of pharmaceutical cold chains need to be managed effectively due to their direct and critical effects on public health.

Pharmaceutical products often require product-specific storage, packaging and transportation conditions that necessitate strict control to ensure their integrity and compliance with regulatory standards. The highest risk of temperature variations occurs during the transfer of the products. Pharmaceutical products are typically transported with refrigerated vehicles to meet the desired temperature during the route. Failure to maintain the desired temperature throughout the route, in other words, breakage of the cold chain, may cause serious health risks. Additionally, breakdown of the cold chain may lead to substantial financial risks both for manufacturers and logistics companies owing to wasted inventories, regulatory penalties, and damaged reputation. Hence, managing pharmaceutical cold chain risks is essential, as their

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consequences can be severe and fatal. Machine learning algorithms are suitable for risk classification and prediction in the pharmaceutical cold chain field (Konovalenko & Ludwig, 2021; Mangini et al., 2024). Outlier detection is one way to handle such risks (Zhao et al., 2013).

Transportation of temperature-sensitive pharmaceuticals is a challenging task that requires monitoring temperature deviations and detecting anomalies that could indicate a cold chain breakage. In order to contribute to this critical research area, this article employs hybrid Route Detection-based Support Vector Regression (RD-SVR) algorithm for outlier detection during transportation of temperature-sensitive pharmaceutical supply chains. The related research question is “Can RD-SVR algorithm be used to detect out-of-reference temperature measurements on location-based temperature data in pharmaceutical cold chain logistics?”.

The RD-SVR model proposes a framework for classification-based outlier detection in cold chain logistics. The proposed model consists of two parts. The first part is the Route Detection (RD) algorithm. This algorithm determines the routes by tracking the location through coordinate measurements and temperature checks of the sensors at the same time. The starting points of the routes are always the operations center. The endpoints of the routes are determined based on a threshold set for the algorithm. Any data beyond the endpoints is eliminated as part of the data cleaning process. Thus, the RD algorithm not only determines routes and cleans data but also provides a dynamic structure for companies. The proposed model is applied on data gathered from two vehicles of an international logistics company for a one-year time horizon in Türkiye. The case study company has ten predefined routes. However, these routes are not always strictly followed and may change based on customer demand over time. Therefore, even if the routes change according to customer requests, the developed RD algorithm detects new routes, ensuring a dynamic and real-time structure. In the second part, classification-based outlier detection is performed using Support Vector Regression (SVR) for each route determined by the RD algorithm to detect anomalous temperature during transportation. The advantages of using SVR can be expressed by its robustness and being convenient to multi-dimensional data (Awad et al., 2015).

The RD-SVR algorithm is tested against Random Forest and ANN. Test results and sensitivity analysis highlight the robustness and success of this innovative classification-based outlier detection model for identifying cold chain breakages. The dynamic route-based structure, which can be considered as a preprocessing task, makes the algorithm highly suitable for logistics companies that require the processing of big data. Moreover, the dynamic route-based structure is also ideal for logistics service providers managing multiple routes and dynamic schedules.

The remainder of the article is structured as follows. Section 2 presents the relevant literature. Section 3 explains the proposed approach, case study company, and data collection. Section 4 gives and discusses the results. Finally, Section 5 concludes the paper.

2. Literature review

2.1. Cold chain risks

Cold chain logistic refers to the post-production supply chain activities of temperature-sensitive products within the required temperature and humidity ranges to guarantee product safety (Joshi et al., 2009; Salin & Nayga, 2003). The main definition of the cold chain can be described as controlling and monitoring the temperature of the related products along the entire supply chain (Shashi, 2018). Cold chain products are categorized in five groups: 1) fruits and vegetables; 2) bakery and confectionary; 3) dairy and frozen dessert; 4) meat and sea food; 5) drugs and pharmaceuticals (Allied Market Research, 2022). This classification can be categorized under two main fields; food cold chain and pharmaceutical cold chain (Centobelli et al., 2021).

A supply chain inherently involves multiple sources of risk. In a supply chain, risk can be defined as failure, damage or loss caused by

unforeseen events (Zsidisin & Ritchie, 2009). There are many risk categorizations of supply chain. Yıldız Özenc et al. (2023) provides a comprehensive literature review which integrates all risks and risk categorizations in a supply chain. Risks in the cold chain are vital and should be kept under control. In literature, cold chain risks have been categorized in many ways. Liao et al. (2023) stated that cold chain risk occurs due to lack of infrastructure, lack of green transportation, lack of capacity building, operation problems in cold chain transport, loss of energy, temperature control and packaging. Shen & Qian (2023) identified five main risks of cold supply chains: refrigeration risk, supply and demand risk, logistics risk, information risk and environmental risk. Lin et al. (2023) developed a hierarchy model for vaccine transportation quality and risk in cold chain with five criteria and twenty-six sub-criteria and determined the most influential sub-criteria for vaccine transportation quality and risk as local weather conditions, topography, road characteristics, cold-chain logistics standardization, and national economic development. Ding et al. (2023) identified fifteen risk factors in three risk assessment aspects, operation risk, hardware risk, and shipper risk, that threaten cold chain operations. They concluded that the most important risk aspect is shipper risk and six key factors that significantly impact cold chain logistics are excessively long container loading time, insufficient pre-cooling of goods, poor or improper product packaging, dockside or shipboard power supply system malfunction, improper cargo stacking, and human error affecting temperature settings by using the best-worst method. Arslan et al. (2023) identified the following risks for food cold chains; financial risks, delivery risks, technological ability risks, environmental risks, quality risks and social risks. They prioritized these risks by using Decomposed Fuzzy Set-Based FMEA approach.

Pharmaceutical cold chain has special regulatory requirements for transportation and storage (Ruiz-Garcia & Lunadei, 2010). Pharmaceutical cold chains face various challenges that can impact quality and efficacy of temperature-sensitive products. In pharmaceutical cold chains, all links of the chain are connected to each other. If there is a breakdown in any one of them, the subsequent steps are also affected (Bamakan et al., 2021). The major challenge in preventing any breakdown is temperature control, ensuring that the temperature remains within the desired range (Bogataj et al., 2005). Product spoilage caused by temperature fluctuations has serious effects on human health. There are also other risks that can cause a cold chain breakdown, such as equipment failure, unreliable electricity power systems, and poor road conditions (Lauton et al., 2019; Li et al., 2024). Cold Chain failures are not only logistic issues. The consequences of these failures can affect directly human health. Mitigating cold chain risks requires a combination of advanced technology, regulatory compliance, trained personnel, and robust logistics strategies. Besides, real-time temperature monitoring and AI-driven risk management can enhance cold chain reliability.

As a result of the literature research conducted to identify cold chain risks, we concluded that every risk in the cold chain indicates one major risk. That risk is the breakage of the cold chain. The most important reason for the breakage of the cold chain is the situation where the temperature goes out of the limit during transportation.

2.2. Outlier detection

Cold chain monitoring involves recording, managing and verification of the temperature indicators/humidity of products as they move through the supply chain. It maintains the temperature-sensitive products, such as pharmaceuticals, chemicals and perishable food, remain within specified environmental conditions. Traceability in the cold chain is crucial for ensuring quality, safety and regulatory compliance. Technology plays a vital role in real-time tracking; e.g., temperature data loggers, RFID-enabled smart labels, IoT-based sensors, GPS and real-time location tracking, cloud systems, smart packaging, and biosensors. Cold chain logistics generate enormous volumes of real-time

data due to these advanced tracking technologies. Advanced analytics, big data platforms, and AI-driven models plays a critical role in transforming this raw data into actionable insights for decision-making and risk management.

Cold chain monitoring heavily relies on outlier detection to ensure reliability. Outlier detection is the process of identifying deviations from expected temperature, humidity, or handling conditions, which could indicate potential risks to product quality and safety. Out of the limit temperatures can be interpreted as outliers in a series of data. Therefore, detecting the breakage of the cold chain can be considered as an outlier detection, also known as anomaly detection, problem. In a way, outliers can be defined as inconsistent values in a set of data (Barnett et al., 1979). There are several methodologies of outlier detection, and they are grouped in different ways in the literature (Escalante, 2005; Han et al., 2012). These methodologies can be categorized as statistical-based techniques, density-based techniques, classification-based techniques, clustering-based techniques, and also machine learning and deep learning techniques (Chandola et al., 2009). Machine learning is a useful approach in an industrial context and has a wide range of applications (Bertolini et al., 2021; Oliveira et al., 2024). Anomaly detection is one of the crucial machine learning tasks (Kang et al., 2020). Numerous studies apply machine learning techniques for anomaly detection in the existing literature (Hernandez-Jaimes et al., 2024; Mascali et al., 2023; Oliveira et al., 2024; Ravinder & Kulkarni, 2023; Ugli Olimov et al., 2022; Zhao et al., 2013). As well as machine learning algorithms, deep learning algorithms have recently become popular for anomaly detection (Abusitta et al., 2023; Seo et al., 2024; Tian et al., 2025; Zipfel et al., 2023).

Various learning algorithms such as artificial neural networks (ANNs), support vector machines (SVMs), Bayesian networks, etc., can be applied as a classification-based outlier detection method (Chandola et al., 2009). SVMs have always been a preferred method in outlier detection for various study fields (Baldomero-Naranjo et al., 2021; Thongkam et al., 2008; Zhang et al., 2013). In the study, where classifiers were compared, it was concluded that these classifier models can distinguish various anomalies, and it was determined that the SVM had the highest predictive power among the classifiers (Carvalho et al., 2019). SVM can be generalized for regression (Alpaydin, 2020). This generalized version is called Support Vector Regression (SVR). SVR is a machine learning method where a model determines the significance of a variable in describing the relationship between input and output (Zhang & O'Donnell, 2020). SVR is used as the classification-based algorithm. Similar to classification, SVR is defined by its use of kernels, a sparse solution, and control over the margin and the number of support vectors (Awad et al., 2015). The classification-based outlier detection methods such as ANN, SVR, and Random Forest for the cold chain are not sufficient for outlier detection their own. Geotagged data from vehicles is extensive and requires thorough cleaning before any classification methods can be applied. There is a gap in the literature in this area.

Outlier detection in pharmaceutical cold chains presents additional challenges compared to standard cold chains, due to the highly sensitive nature of products such as vaccines and temperature-sensitive medications. Pharmaceutical cold chains face numerous challenges: i) Many pharmaceutical products require strict temperature intervals and precise temperature control (often between 2–8 °C or even colder). ii) Even small temperature deviations can have significant effects on the quality and safety of the products. These effects may include degradation, reduced effectiveness or becoming harmful for human health. iii) Pharmaceutical cold chains may span multiple regions; hence, the geographical diversity may increase the risks due to differences in seasons, environmental factors and regulations. iv) Pharmaceutical cold chains require compliance with stringent processes, regulations, and audits. Traditional approaches are often based on basic statistical techniques or simple thresholds. However, real time visibility and advanced outlier detection systems such as machine learning-based models are

required to handle the growing volume, variety and velocity of modern pharmaceutical cold chains. Advanced AI-based models may help to detect non-obvious changes earlier or provide predictive risk analytics. Moreover, pharmaceutical transportation vehicles mostly include GPS-enabled data loggers which continuously record temperature and location data. This data provides an opportunity to link outliers to specific locations or perform route optimization based on historical anomaly trends.

3. Methodology

This section explains the research methodology of this research study under the concept of objectives, research questions, techniques, and work plan. Based on our research question, classification-based outlier detection framework is developed as route detection-based support vector regression algorithm. Steps of the framework are planned as research planning, literature review, data collection, data mining analysis, and results. The flow chart of the developed framework is given in Fig. 1.

In Phase I, meetings, focus group discussions, and field study were conducted to identify the risk related problems and cold chain risks in the collaborating company. Among the identified risks, the most critical risk is the cold chain breakage during transportation. The objective is to identify the anomalous temperature in pharmaceutical cold chain logistics which may cause the risks and damage the product safety. In light of the objective, the following research question is conducted: “Can RD-SVR algorithm be used to detect out-of-reference temperature measurements on location-based temperature data in pharmaceutical cold chain logistics?”

In Phase II, a comprehensive literature review was conducted to identify the most critical cold chain risks in the literature. The most addressed risk is addressed as cold chain breakage both based on literature review and discussions with the company. Therefore, this vital risk was selected as the target problem. In literature, various learning algorithms such as ANN and SVM have been used for outlier detection in cold chains. However, these algorithms alone are not sufficient to address our specific problem. The main objective of the study is to perform route-based outlier detection while providing adaptive structure for varying routes during transportation and robust data preprocessing. The insufficiency of existing methods has driven us to develop a hybrid algorithm that first preprocesses to identify routes and conduct cleaning in large-scale data and then performs outlier detection for each route separately.

Phase III is the data collection phase. The collaborating company provides cold chain transportation data of two refrigerated vehicles over one year period. The company is engaged in the storage and transportation of pharmaceutical products. The cold chain structure is illustrated in Fig. 2.

The company's operation center is located in Kocaeli, Türkiye. Cold products are stored in cold rooms of the warehouse in the operation center. Then the cold products are transported to pharmaceutical warehouses by refrigerated vehicles according to the demand. From pharmaceutical warehouses, cold products are delivered to customers. The temperature is constantly monitored both in the storage area and in transportation by data loggers. The provided temperature data covers data loggers of the vehicles from operation center to pharmaceutical products during transportation over one-year period. Further information about the data is given in Sub-section 3.1.

Phase IV is the analysis phase. Detecting anomalous temperature in cold chain logistics can be considered as a binary classification problem. A classification-based framework is proposed to detect anomalous temperature along the route by developing a route detection-based support vector regression algorithm (RD-SVR). In this phase of the study, the developed RD-SVR model was implemented, parameter optimization was performed and model performance was evaluated along with a comparison with Random Forest and ANN algorithms. In Phase V, the results of the analyses are presented and discussed in the

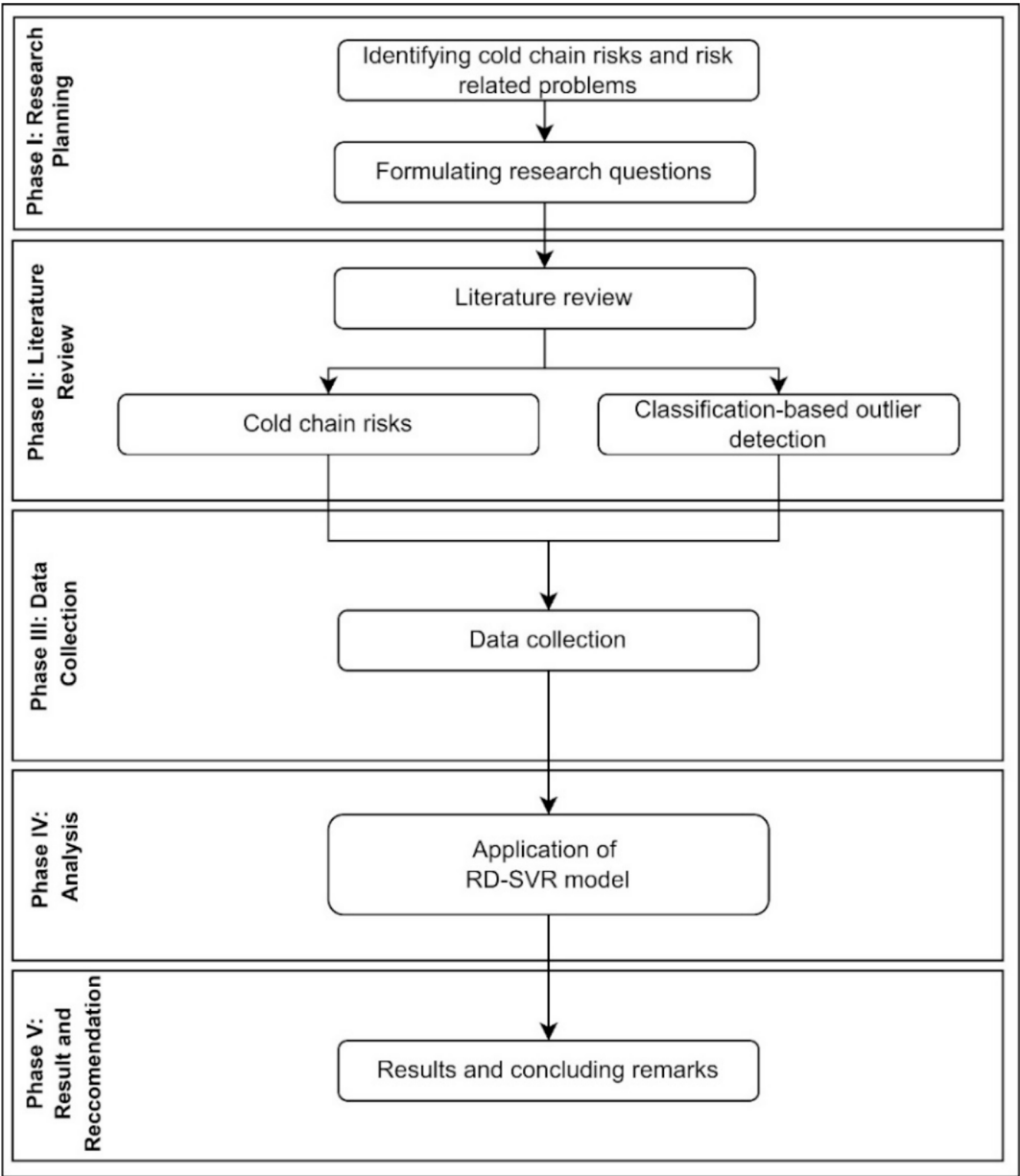


Fig. 1. Flowchart of Research Methodology.



Fig. 2. Cold chain structure of the case study company.

context of robustness, efficiency, and practical contribution of the algorithm.

3.1. Cold chain data

The quality and safety of cold chain products must be ensured from the producer to the end consumer during storage and transportation.

Pharmaceutical products are transported in the cold chain with refrigerated vehicles in the cooperated company. When transporting products, the temperature value in the vehicle should be maintained in desired range. The company's refrigerated vehicles are cooled with air conditioners. There are two air conditioners in each vehicle in case of any breakdowns. Monitoring the temperature and humidity inside vehicles is carried out using data loggers. These devices are used to monitor the internal environmental conditions of vehicles and collect data.

The case study company provides storage and transportation services for pharmaceutical cold chains. They use top-notch technology in both storage and transportation to keep the products in the desired temperature range. The desired temperature range is between 2 °C and 8 °C. The air conditioners keep the temperature in the desired range. The situation in the storage area is more stable and under control. In case of any malfunction, backup air conditioners are activated, and the technical team intervenes immediately. For this reason, out-of-limit situations are rarely experienced in the storage area. However, situation in transportation is a bit challenging since the Türkiye's road characteristics has some difficulties. Air conditioners fail more often. Consequently, we chose to investigate the transportation in cold chain rather than storage as a research area.

On the company side, data quality is ensured by using two simultaneous data loggers which have temperature sensors. For the measurements, both loggers act as a substitute for each other. If one fails to measure, the other one provides the measurement. Unfortunately, there can be missing data if both sensors fail at the same time. In addition, simultaneous measurement is used to ensure the accuracy of the measurement. Also, the measured data is transferred to a remote database and saved locally. This local save is a precaution to prevent possible data loss during any occasional connectivity problems.

The input data is the data saved locally by the loggers. The accuracy of each measurement was determined by comparing two simultaneous readings and no different values were detected. Hence, it is accepted that there is no missing and inaccurate value in the input data set. The data set consists of measurements done in two-minutes intervals. Although this interval is dominant in most of the data, we examined that the time difference between two consecutive measurements may vary. This situation could be considered potential data loss. Due to the lack of information on this matter and the fact that both sensors adapt to the changing measurement interval, we do not classify it as data loss.

The company provided one year of data from the data loggers in the two vehicles on route base. Data loggers are measuring devices used in temperature and humidity measurement, which record the temperature or temperature-humidity values of the environment in which it is located until its capacity is full, can store the recorded information in its memory and allow the information in its memory to be transferred to the computer when necessary. There are three data loggers in each refrigerated vehicle of the company. Two of them measure the temperature inside the vehicle and one of them measure the humidity inside the vehicle. Each data logger measures approximately every 2 min. Thus, in the data, there are two temperature values, one humidity value, vehicle identity, date, time, latitude and longitude information between April 2023 and March 2024. Humidity values were excluded on purpose, since no action is taken by the company when the humidity limit exceeds 80 %. The customer is only informed during delivery about the humidity issue. An example of the given data, including the headers and tails of a single file is presented in Table 1.

In the data set, there are 3 "type" values in the first column which are TYPE 2, TYPE 2 and TYPE 5. These values are data logger records. TYPE 2 and TYPE 3 represent the temperature values and TYPE 5 represent the humidity values inside the vehicle. In the second column, "value" represents the temperature and humidity values. In third column, "unitid" represents the vehicle identity. The fourth column shows the date and time of the records. The fifth and sixth columns show the coordinate of the vehicle. In this study, TYPE 2 values are used for analysis as

Table 1

An example of the data given.

TYPE	VALUE	Unitid	Date and Time	Latitude	Longitude
2	27	54,808	2023-08-01 00:01:10	40,984723	29,251308
5	59	54,808	2023-08-01 00:01:10	40,984723	29,251308
3	26,799999	54,808	2023-08-01 00:01:10	40,984723	29,251308
2	26,9	54,808	2023-08-01 00:03:10	40,984723	29,251308
5	59	54,808	2023-08-01 00:03:10	40,984723	29,251308
3	26,799999	54,808	2023-08-01 00:03:10	40,984723	29,251308
2	26,9	54,808	2023-08-01 00:05:11	40,984723	29,251308
5	59	54,808	2023-08-01 00:05:11	40,984723	29,251308
3	26,799999	54,808	2023-08-01 00:05:11	40,984723	29,251308
...	...	...	...	...	...
2	29,099999	54,808	2023-08-30 09:38:23	40,881196	29,260263
5	70	54,808	2023-08-30 09:38:23	40,881196	29,260263
3	29	54,808	2023-08-30 09:38:23	40,881196	29,260263
2	29,099999	54,808	2023-08-30 10:10:26	40,881196	29,260263
5	70	54,808	2023-08-30 10:10:26	40,881196	29,260263
3	29	54,808	2023-08-30 10:10:26	40,881196	29,260263
2	29,2	54,808	2023-08-30 10:12:27	40,881196	29,260263
5	70	54,808	2023-08-30 10:12:27	40,881196	29,260263
3	29,099999	54,808	2023-08-30 10:12:27	40,881196	29,260263

temperature data. In-vehicle temperature data from TYPE 2 for August and December (in order to see the differences between summer and winter months) are given in Fig. 3 and Fig. 4. The red lines represent the temperature range which is desired to be within all temperature values. The blue dots demonstrate the temperature values.

Transportation data was gathered as a monthly basis for two refrigerated vehicles over one year period. Each refrigerated vehicle has 12 data files monthly. Each data file has approximately 150,000 columns.

3.2. RD-SVR model

A hybrid approach, Route Detection-based Support Vector Regression (RD-SVR) was developed for classification-based outlier detection for pharmaceutical cold chain to detect anomalous temperature during transportation of pharmaceutical products. The proposed model consists of two consecutive algorithms: Route Detection (RD) algorithm and Support Vector Regression (SVR) algorithm. The route detection algorithm is developed entirely based on the unique cold chain data on hand. In addition to determining the routes, the algorithm also performs big data cleaning, which is one of the most important processes for sensor data. Then, the support vector regression algorithm determines the outliers based on each route during transportation as a classification-based outlier detection method. The flowchart of the proposed RD-SVR model is demonstrated in Fig. 5.

The RD algorithm also provides robust data preprocessing. While determining the routes, the algorithm simultaneously performs data cleaning. Once the routes are defined, unnecessary data is removed. For example, since the vehicle is empty on the return journey, the temperature data measured during the return trip is considered as unnecessary

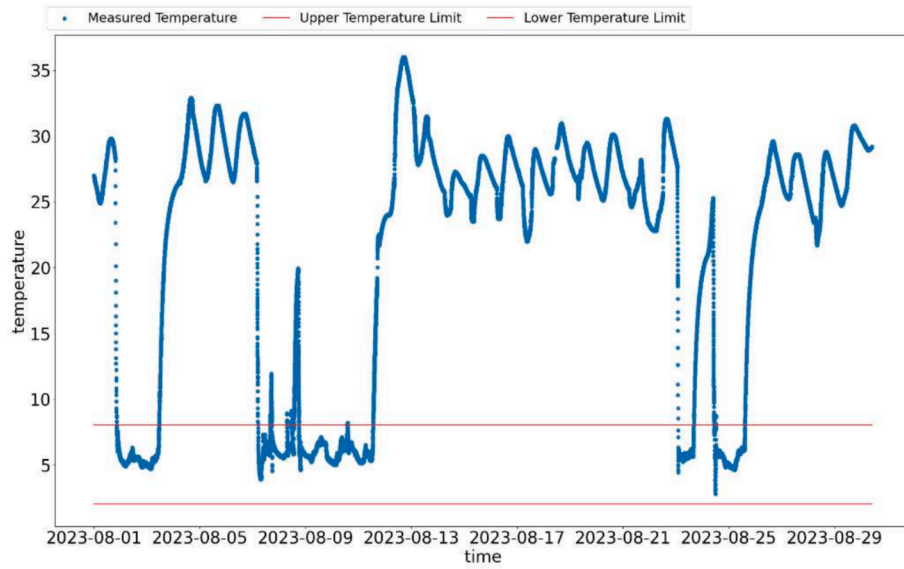


Fig. 3. Temperature vs Time Graph for August.

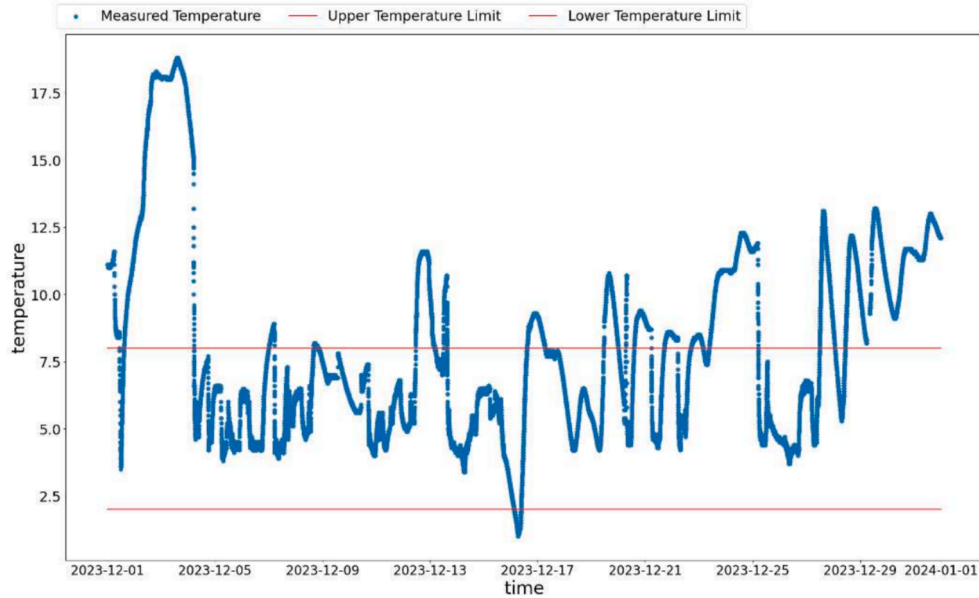


Fig. 4. Temperature vs Time Graph for December.

data. Therefore, the RD algorithm cleans this data, effectively handling preprocessing as well. Additionally, due to the specific requirements of the collaborating company, outlier detection is required to be performed on a route-based approach. Hence, a hybrid algorithm, RD-SVR, has been developed that performs route detection and outlier detection at the same time. This novel approach aims to fill the gap in existing literature for cases where data elimination and route-based features are essential.

3.2.1. Route detection algorithm

Sensor data is considered big data. It is very critical to handle big data cleaning in order to accurately identify anomalies in the field of pharmaceutical cold chain. For this purpose, a rule-based algorithm named route detection is developed to identify the routes while pruning the data. RD algorithm analyses the overall 12 months of data row by row. It aims to identify each distinct route that the refrigerated vehicle takes, by tracking the change in the location and temperature

measurements. This process helps to prune the data by identifying the routes and neglecting the measurement taken when vehicle is inactive.

Reverse geocoding is an essential process to determine the routes of refrigerated vehicles. Location information of the refrigerated vehicles is given as latitude and longitude information in the data. However, to determine routes, the address of the vehicle is needed to know instead of the coordinates. Therefore, reverse geocoding is employed to transform latitude and longitude to the address.

In order to apply reverse geocoding, all latitude and longitude pairs of each vehicle from different data files are combined into a set of coordinates. To accomplish this, only a unique set of coordinates can be worked on. Eventually, approximately 300,000 unique coordinates data remained. A Python script is prepared, and reverse geocoding service is used to convert the data. The script communicates with the service by sending a coordinate and writing the result back into a file.

As we know, the route of the vehicles always starts with the operation management center of the company, which is located in Kocaeli,

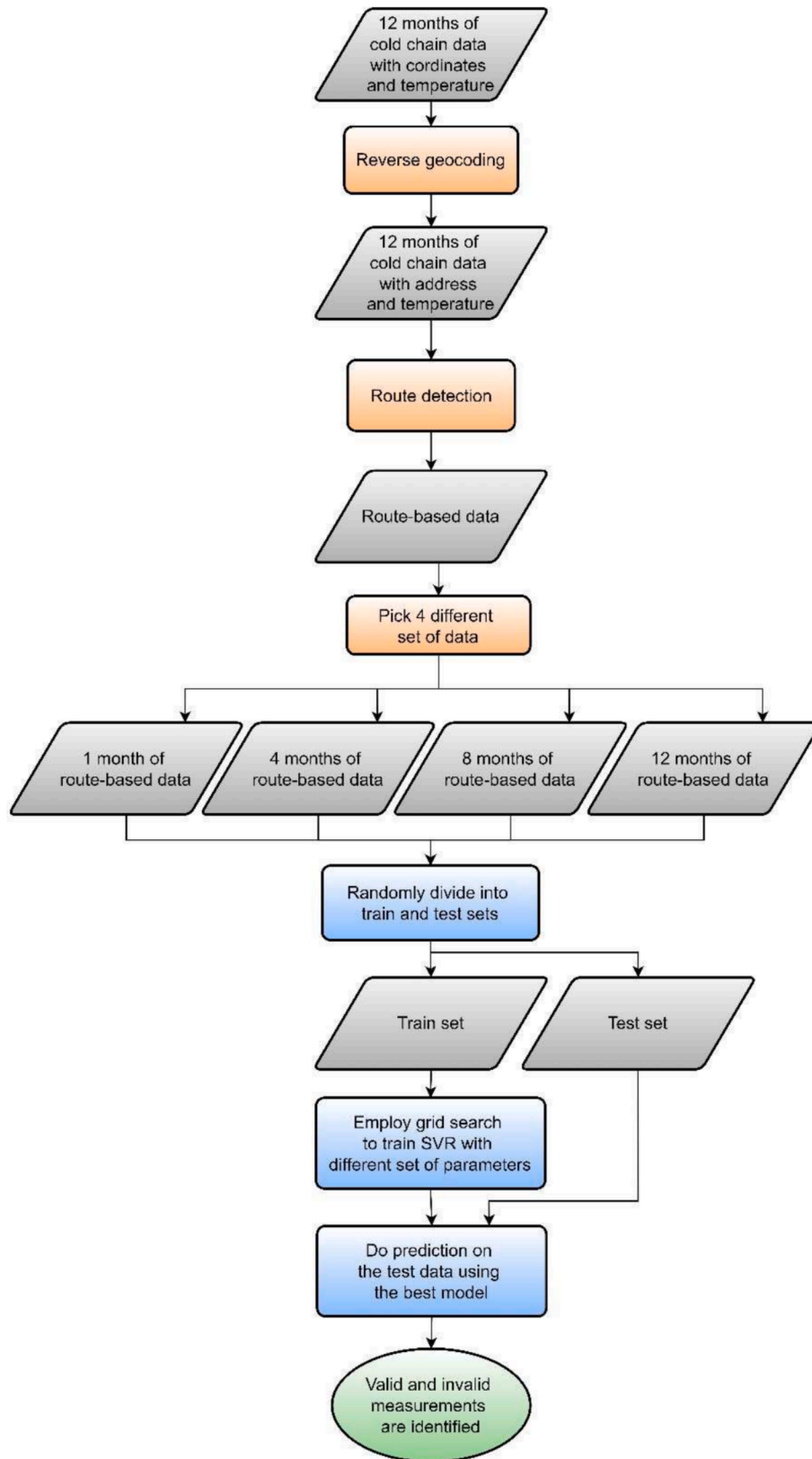


Fig. 5. Flowchart of RD-SVR Model.

Türkiye. The end point depends on customer demand. Since the final destination may vary, the routes of the vehicles should be extracted from the data. To find the routes, we checked the changing addresses starting from Kocaeli in the data on a city-by-city basis by considering

temperature inside the vehicle. Each vehicle must maintain an inside temperature between 2 °C and 8 °C along the route. The cooling inside the vehicles begins before loading. Thus, the interior temperature values during loading meet the required temperature values. We set the

threshold as the interior temperature of the vehicle meets the 8 °C or under very first time while the vehicle is at Kocaeli location. Then, we eliminated the data before this threshold.

The key aspect of the route detection algorithm is to determine the city where the vehicle finishes unloading. Because the final destination of the routes can be changed according to customer demand. The vehicles should be active on the routes, and they do not have to be active after delivering the pharmaceutical products to the final destination. In other words, the air conditioners are switched off on the way back and the temperature inside the vehicle starts to increase. Even if the vehicles are inactive data loggers inside the vehicles record the temperature and humidity in almost two minutes. Therefore, the data on when the vehicles are inactive should be eliminated. While eliminating the data, we examined the cumulative increase in temperature and set a threshold for the temperature. A cumulative temperature increase of 2 °C is considered highly manageable. If a smaller increment is chosen, it may lead to the elimination of some outliers. Choosing a larger increment may yield a greater number of unnecessary measurements in the data. The selection of thresholds and the model's performance at different thresholds are detailed in Section 4.3. A cumulative increase of 2 degrees is considered suitable for this study. When the temperature reached 10 °C, we assumed that the vehicle was on the way for returning to the operation center. Temperature data between 8 °C and 10 °C on the return should not be considered as an outlier. Besides, these data constitute a very small part of the whole data. The data higher than 10 °C on the way returning is eliminated by the developed RD algorithm. Like this way the RD algorithm prunes the data. In the data cleaning process using the RD algorithm, temperature data was examined and no missing values for temperature measurements were detected. Additionally, in most of the data, measurements are taken every 2 min, but occasional variations in this interval, such as 3 or 5 min, are not considered sensor errors. The threshold is used to retain outliers and unnecessary data is eliminated with the algorithm. As illustrated in Fig. 6, most of the unnecessary data was eliminated by using the RD algorithm.

All routes over one year period are determined monthly for each vehicle by using the developed RD algorithm. If an extreme situation occurs on routes, such as air conditioning failure, the vehicle door being left open, or an accident on the road, it is assumed that this will be reflected in the data. If the extreme situation does not affect the internal temperature of the vehicle, it is assumed that the cold products remain undamaged. However, if the internal temperature exceeds the limit, these extreme situations are reflected in the temperature data, and the

algorithm classifies them as outliers. The routes of the vehicles which took six routes in August are given in Fig. 7 as an example.

All the routes of each vehicle are demonstrated on the maps of Türkiye. As an example, Route 2 of the vehicle in August is given in Fig. 8. The route is stated as a green line on the map. Coordinates of the vehicle with valid temperature are stated as a blue pin and coordinates of the vehicle with invalid temperature stated as a red pin on the map in Fig. 9. Fig. 10 demonstrates the route after data elimination as a part of the RD algorithm with valid and invalid temperature. These figures were created as HTML files. When the cursor hovers on the pins, it shows the temperature, date and time of the measurements. A map-based representation of all routes can be shared upon request.

3.2.2. Support vector regression

If the training dataset has class labels, outlier detection can be considered as a classification problem and any classification methods can be used (Han et al., 2012). In our problem, the class label is very clear: Valid and invalid temperature. Support Vector Machine (SVM) is selected as a classification method for outlier detection in this study. The SVM can be used in several ways in binary classification. As a variation of SVM, we used Support Vector Regression (SVR) which is proposed by (Drucker et al., 1996) and used only for classification problems (Fadil et al., 2021). SVR is a data mining algorithm used to predict a real number by mapping an input data to a function, using training data (Yu & Kim, 2012). SVR has been a preferred method for anomaly detection and real time monitoring (Kromanis & Kripakaran, 2013; Lee et al., 2020). Based on a theoretical principles and empirical studies, SVR offers robustness to outliers on the data. It supports kernels to detect nonlinear relationships in the data. Additionally, unlike other methods such as Random Forest and Decision Tree which has a risk of overfitting on small amount of training data, SVR can perform well on limited data. To match the requirement for SVR, we labeled our data as 1 for valid and 0 for invalid temperature values.

SVR can be used with custom and predefined kernels which have several parameters. In context of this study, we tested the linear, polynomial, Radial Basis Function (RBF) and sigmoid kernels with various c and γ values. The parameter c determines the cost of the mislabeled predictions. Picking a higher c value helps the model to accurately classify all training examples. The γ parameter determines the significance of a single training example. Picking a higher γ means that other examples need to be closer to have an impact.

In this study, the SVR implementation which is provided by Sckit-

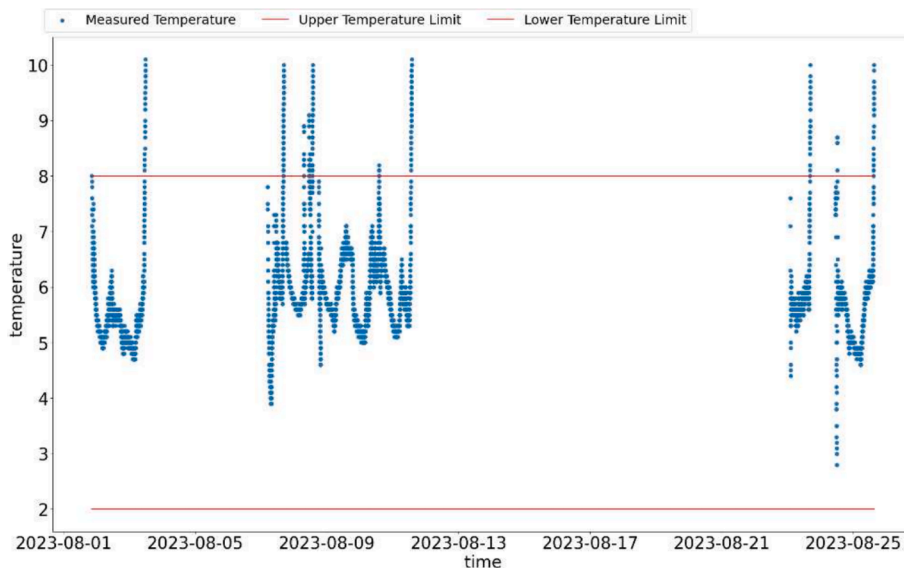


Fig. 6. Temperature vs Time for August after data pruning with the RD algorithm.

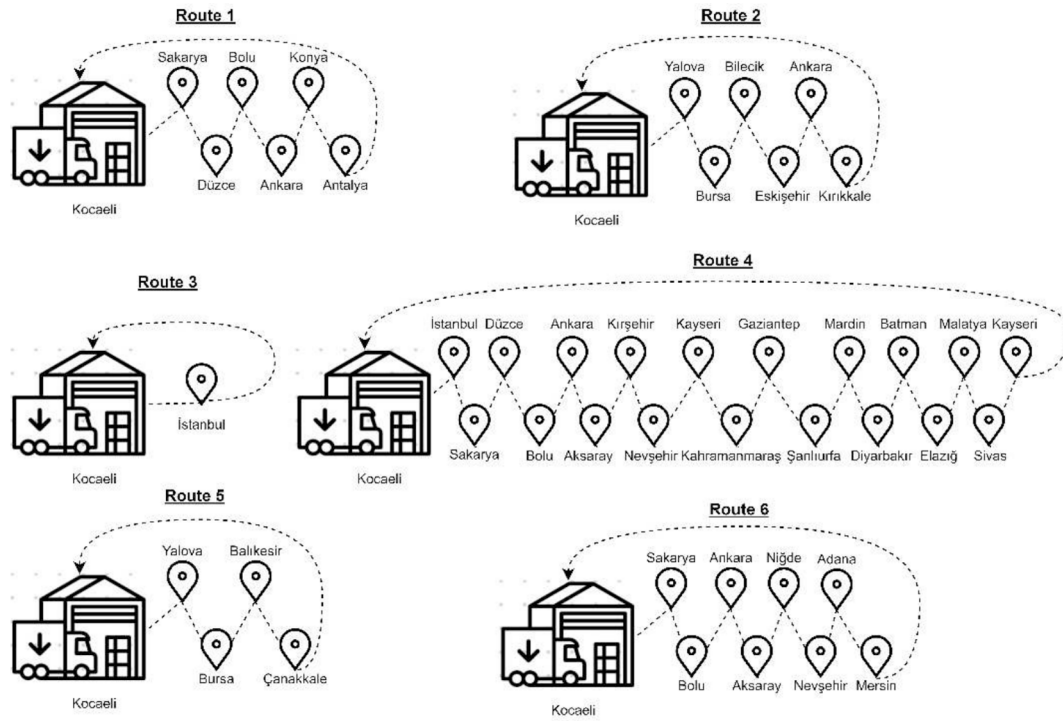


Fig. 7. All routes of a refrigerated vehicle in August.

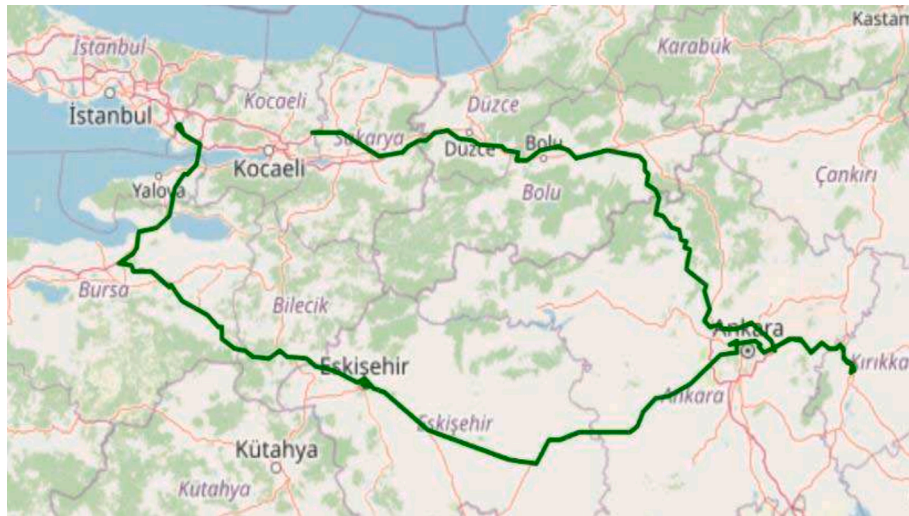


Fig. 8. Map-based representation of Route 2 in August.

learn is used (Buitinck et al., 2013; Pedregosa et al., 2011). The default values of the parameters in this library are RBF for kernel, scale for gamma and 1 for c. Also, hyperparameter optimization was conducted to find the best kernel along with the best set of parameters. Hyperparameter optimization details, values for each parameter and results are given in Section 4.

4. Results

4.1. RD-SVR results

Hyper parameter optimization is used to select the optimal parameters (Claesen & De Moor, 2015; Yu & Zhu, 2020). Here, the terms parameters include the kernel function, c and gamma values. Possible sets of parameters are processed through grid search. The sets of all

parameters are given in Table 2.

The cross-validation value for the grid search is taken as 3 and the candidate sets of parameters and scored based on Means Squared Error (MSE).

A total of 12 months of data was given to the algorithm as 4 different data sets. The data sets contain 1-month, 4-months, 8-months and 12-months of data. The reason for giving data to the algorithm as four different data sets is to investigate the performance of the developed RD-SVR model on different data sizes. Each data set is randomly split into two parts, 20 % test and 80 % train. For each train set, grid search returns the parameter set which gives the best result. The best set for each run is given in Table 3.

For each set of parameters determined by the grid search, R^2 score and MSE are calculated using the test sets. Results are given in Table 4.

According to the results, SVR provides high accuracy based on a high

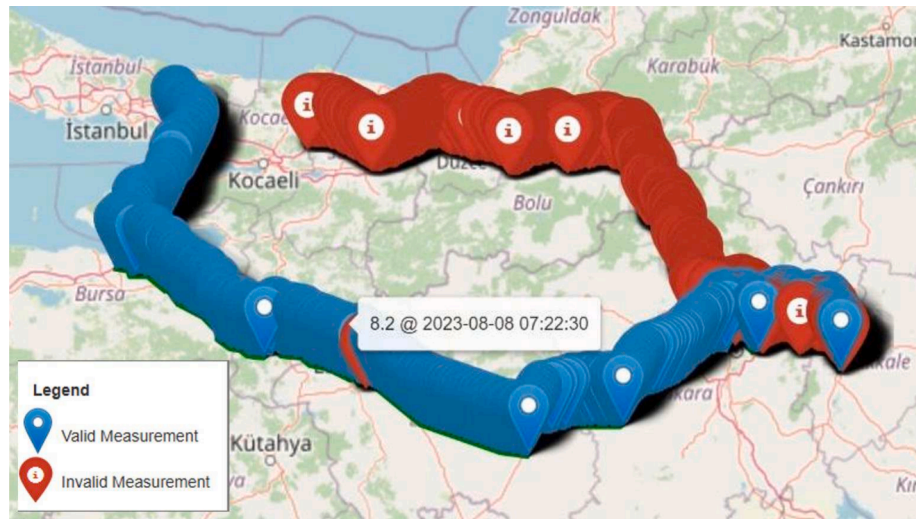


Fig. 9. Map-based representation of Route 2 in August with location pins for each temperature measurement.

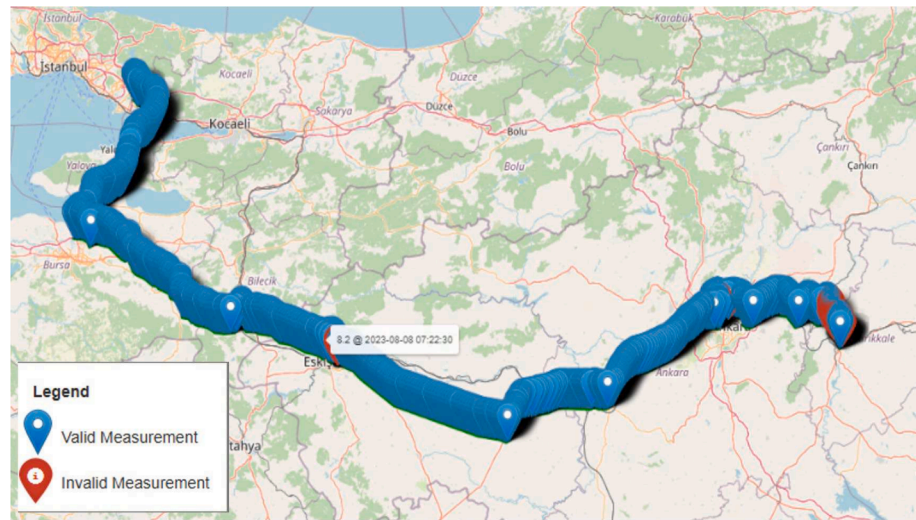


Fig. 10. Map-based representation of Route 2 in August with location pins for each temperature measurement after data elimination with the RD algorithm.

Table 2
Parameter values for SVR.

Parameter Name	Parameter Value			
Kernel	linear	polynomial	rbf	sigmoid
C	0.1	1	10	100
gamma	auto		scale	

Table 3
Optimal set of parameters for each data set.

Data sets	C	Kernel	Gamma
1-month	100	rbf	auto
4-months	100	rbf	scale
8-months	100	rbf	scale
12-months	100	rbf	scale

R^2 value and low error terms, indicating that the SVR algorithm performs well for this study. The fitted graph of the SVR predictions for all data sets is given in Fig. 11. According to the graphs, for both valid and invalid samples, the predictions from SVR align with the training data.

Table 4
Results for each optimal set of parameters.

Data set	Parameters	R^2	MSE
1-month	c:100, kernel:rbf, gamma:auto	0.7958	0.0039
4-months	c:100, kernel:rbf, gamma:scale	0.9183	0.0084
8-months	c:100, kernel:rbf, gamma:scale	0.9136	0.0077
12-months	c:100, kernel:rbf, gamma:scale	0.9174	0.0073

Classification results of the RD-SVR model for all data sets are given as confusion matrix in Fig. 12.

Anomalous temperature detection using proposed RD-SVR model results for each data set are given in Table 5. The table shows Precision, Recall, F1-Score and Accuracy value for each label in the data sets. In the table, label represented as 0 and 1 which indicates invalid temperature and valid temperature, and support represents the number of data point of each label in each data set. The results of the data sets are compared according to Precision, Recall, F1-Score and Accuracy. As the data size increases, precision, recall and F1-score of anomalous temperature is getting higher which indicates that the RD-SVR model is suitable for big data. Additionally, the accuracy of the model is high for each data set.

This paper also evaluates the performance of the developed RD-SVR

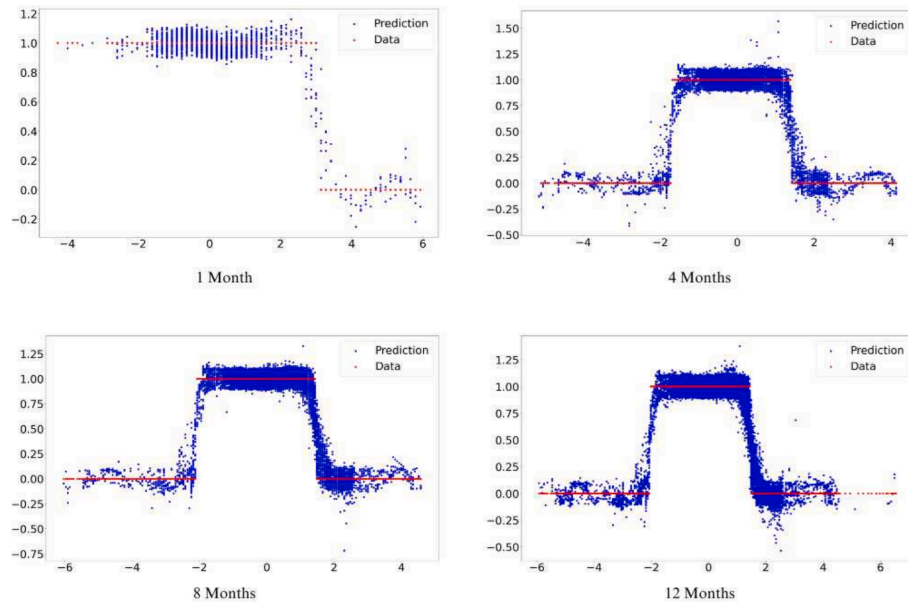


Fig. 11. Visual representation of SVR's train set and predictions for all data sets.

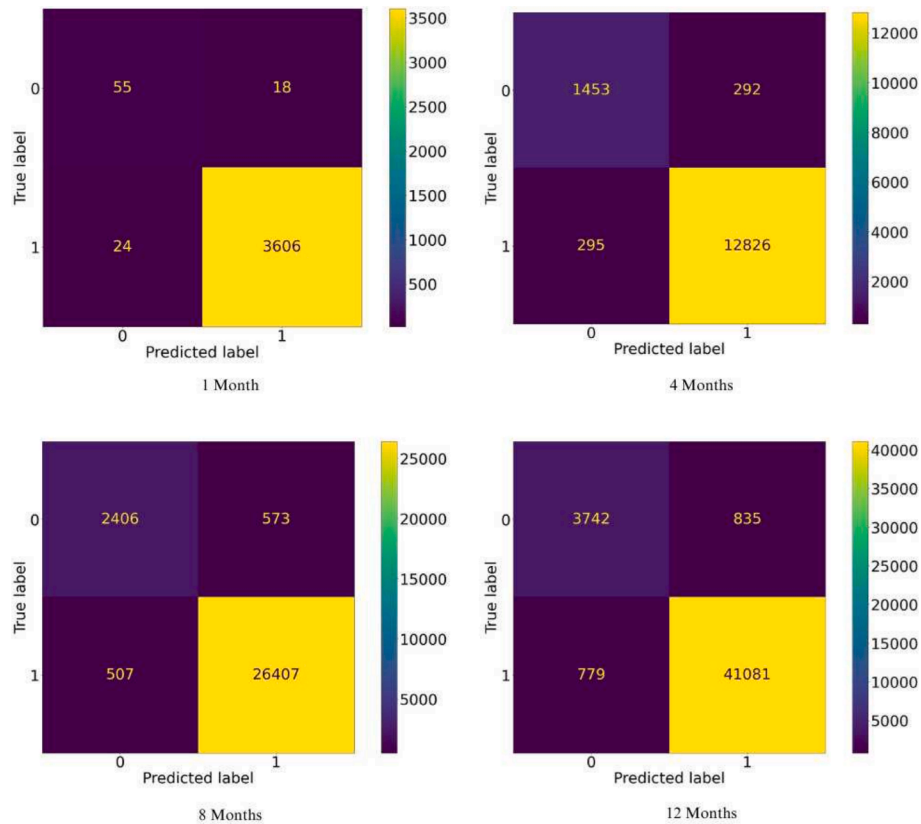


Fig. 12. Confusion matrices for all data sets.

model for different data sizes. As indicated in Fig. 12 and Table 5, our proposed hybrid approach performs well for each data set.

4.2. Comparison

Accurately predicting anomalous temperature in pharmaceutical cold chain logistics is crucial to maintain product safety. In this study, a comparison was made with other machine learning algorithms such as

Random Forest (RF) and Artificial Neural Network (ANN) to test the performance of the RD-SVR model, which was developed for detecting anomalous temperature values. These algorithms were modeled with various sets of parameters. The core parameters of RF are the number of trees in the forest selected as 1, 50 and 150, maximum depth of each tree selected as none and 5, minimum number of samples required to split a node selected as 5 and 10 and minimum number of samples required to be in a leaf node selected as 5 and 10 arbitrarily. The parameters of ANN

Table 5

RD-SVR models' results for all data sets.

Data Set	Label	Precision	Recall	F1-Score	Support	Accuracy
1 Month	0	0.70	0.75	0.72	73	0.99
	1	1.00	0.99	0.99	3630	
4 Months	0	0.83	0.83	0.83	1745	0.96
	1	0.98	0.98	0.98	13,121	
8 Months	0	0.83	0.81	0.82	2979	0.96
	1	0.98	0.98	0.98	26,914	
12 Months	0	0.83	0.82	0.82	4577	0.97
	1	0.98	0.98	0.98	41,860	

are number of hidden layers selected as 1 and 2, size of hidden layers selected as 8, 16 and 32 and activation function selected as Sigmoid and ReLU arbitrarily. The RF and ANN algorithms were applied to the raw data with the selected parameter combinations. Additionally, the RF and ANN algorithms were combined with the route detection algorithm. Then, RF, ANN, Route Detection-based Random Forest (RD-RF) and Route Detection-based Artificial Neural Network (RD-ANN) were compared with the RD-SVR model. The performance metrics of these five models are presented in Table 6.

The models were compared in terms of precision, recall, F1-score, and accuracy. Table 6 demonstrates the best models with high accuracy of each algorithm within variety of parameter combinations. In the table, both the RF and RD-RF models produced the same results. Although these models seem to provide the highest accuracy, both may be overfit since they gave the same results for all parameter combinations. In this case, the results of the RF and RD-RF algorithms are not reliable and not suitable for this study. ANN and RD-ANN algorithms provide better results at some points such as F1-score. However, the precision of predicting outliers is not better than RD-SVR model. Additionally, both ANN and RD-ANN algorithms behave differently for other parameter combinations. For this reason, both models are not consistent. However, a well-modeled ANN algorithm can be a promising research area for the future direction of outlier detection in the cold chain. On the other hand, the RD-SVR model performs well with high accuracy for all parameter combinations. This makes the model more reliable and robust than others. Finally, there are many values in the raw temperature data that behave like outliers even though they are not outliers. These values are caused by the air conditioners turning off when the vehicles begin their return journey after making the last delivery. Even though there are no cold products inside the vehicle, the data loggers continue to measure temperatures on the return journey. As a result, nearly half of the data is labeled as outliers. The developed RD algorithm handles this problem precisely. The algorithm first determines the vehicle's route. Then, by detecting the endpoint of the route, it ensures that temperature values measured after that point are cleaned. Therefore, on their own, RF and ANN algorithms are not very reliable in detecting outliers. These results emphasize the importance of RD algorithm for detecting outliers on route-basis.

Table 6

Comparison of RD-SVR model's result with other machine learning algorithms.

Model	Label	Precision	Recall	F1-Score	Support	Accuracy
RF	0	1	1	1	57,139	1
	1	1	1	1	44,452	
ANN	0	0.80	1	0.89	57,139	0.86
	1	1	0.68	0.81	44,452	
RD-RF	0	1	1	1	4577	1
	1	1	1	1	41,860	
RD-ANN	0	0.80	1	0.89	4577	0.97
	1	1	0.97	0.99	41,860	
RD-SVR	0	0.83	0.82	0.82	4577	0.97
	1	0.98	0.98	0.98	41,860	

4.3. Sensitivity analysis

The RD-SVR model's robustness and reliability were assessed under different conditions. A sensitivity analysis was conducted for three different scenarios by varying threshold, adding noise to temperature data and making cross-validation with different subsets of data.

4.3.1. Varying threshold

Identifying a threshold is a key aspect of the route detection algorithm to determine end of the route. The decision to determine the routes of the vehicle is based on real-time temperature data. As is well known, regardless of the route length, vehicles must maintain the temperature limits along the route. Once the route is completed, all deliveries have been made, so the vehicle is empty, and since the air conditioning is turned off, the interior temperature starts to rise. When the temperature begins to cumulatively increase and exceeds the maximum limit of 8 °C, the algorithm determines the temperature value to end the route. This temperature was set to 10 °C, as it represents a cumulative increase of 2 °C after the limit is exceeded. The reason for setting the threshold to 10 °C is that a + 2°C increase is easily tolerable. For instance, if the threshold were set to 8 °C, any limit breach would make the algorithm think the route was completed, even if it was in the middle of the route. In this case, both the routes would be determined incorrectly, and potential outliers would have been excluded. On the other hand, selecting a higher threshold would lead to more data behaving like outliers in our dataset, which would affect the algorithm's learning. Fig. 13 demonstrates the amount of data behaving like outliers for different thresholds over one month period.

The threshold was set to four different values to test the model's reliability. Standard deviation of temperature over one year period is 7.9. The thresholds are determined based on the standard deviation. Four random thresholds were selected between the upper limit and adding standard deviation to upper limit which is 8 °C and 16 °C. Therefore, thresholds are set to 9 °C, 10 °C, 12 °C and 16 °C arbitrarily. The RD-SVR model is recreated based on these thresholds. Four models were compared according to their accuracies. Comparison results of different thresholds levels given in Table 7.

The model that provides the highest accuracy is RD-SVR with threshold 10. As expected, increasing the threshold decreases the algorithm's accuracy. The lowest accuracy is observed at RD-SVR with threshold 9. Among the data eliminated at threshold 9, there may also be outliers. Cleaning the data after a cumulative + 1 degree may not be sufficient to capture all outliers which may lead to the low accuracy.

4.3.2. Adding noise

Adding noise to a data set is a common method to test robustness and generalizability of the model in machine learning. This approach helps to simulate real life imperfection in data, enhancing the model's ability to handle small variations. Gaussian noise was added to the time series temperature data to test the model's resilience by adding random perturbations to the data set. Introducing small amounts of Gaussian noise to the temperature data helps to evaluate if the detected outliers remain consistent. The added noise follows a Gaussian distribution with a mean of 10 and a standard deviation for the noise values to be greater than 8 °C in temperature readings, affecting 10 % of the total data over one year period. As a result, the RD-SVR model was successful in detecting 66 % of the noise. The model's success in detecting the noise enhances its robustness.

4.3.3. Cross-validation with different subsets of data

The RD-SVR model's performance in terms of seasonality was assessed by dividing the data set into different time periods, such as winter and summer to check whether outlier detection remains consistent. The purpose of this was to determine whether the model could handle certain challenges specific to Turkey. For example, distributing cold products to the Eastern Anatolia region during winter can be

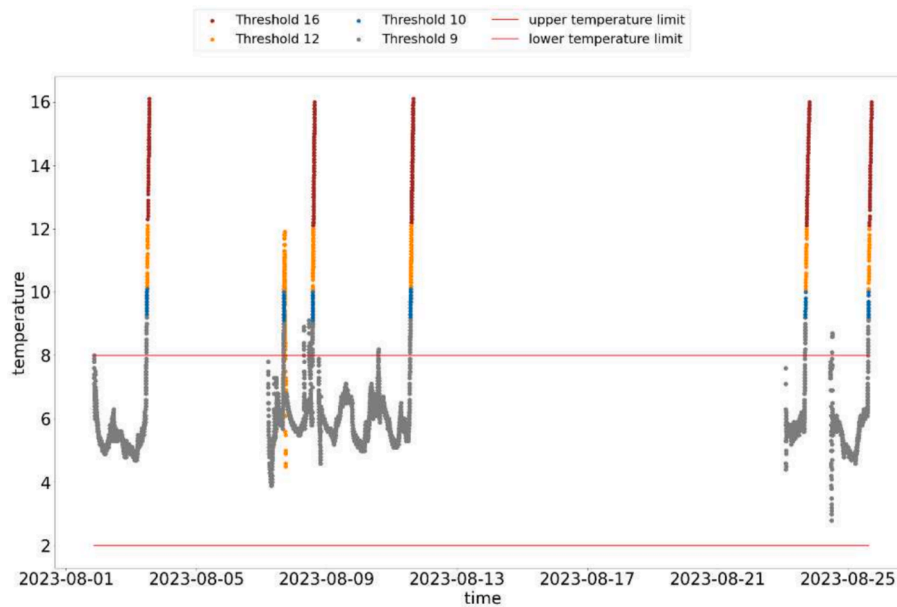


Fig. 13. Temperature vs Time graph for various threshold levels.

Table 7
Comparison of RD-SVR models with varying thresholds.

Model	Accuracy
RD-SVR with threshold 9	0.49
RD-SVR with threshold 10	0.97
RD-SVR with threshold 12	0.81
RD-SVR with threshold 16	0.69

challenging due to the region's rugged roads and extremely cold, snowy weather conditions. Vehicle breakdowns are more common during these periods. Another example is distribution in the Mediterranean region during summer, where extreme temperatures can reach 40 °C or more. This makes it difficult to maintain the in-vehicle temperature at a maximum of 8 °C. Extra care must be taken during deliveries in such conditions.

The data set was divided into three-month periods to assess the model's resilience across different time periods: December, January, and February as winter & June, July, and August as summer. RD-SVR with 3-fold cross-validation was applied to analyze the impact of summer and winter seasons. The model's performance was compared based on the accuracy level. The accuracy of the winter model was 0.56 and the accuracy of the summer model was 0.47. Although the performance of the RD-SVR model train on winter and summer data appears low in terms of accuracy, this is because of the data size is relatively much smaller than the total data and the data sets have same characteristics. Despite their lower accuracy, both models have high R^2 score which are 0.92 and 0.87 respectively and low MSEs which are 0.008 and 0.004 respectively.

5. Discussion and conclusion

The cold chain is an important part of global trade. Any failure to transport cold products in a healthy and safe way affects the links in the supply chain, whether companies or customers. Therefore, cold chain storage and transportation involve significant risks, and the effects of these risks are substantial as well. Pharmaceutical transportation, one of the most crucial aspects of the cold chain, uniquely involves both regulatory risks and human health risks. Due to these risks, a more careful approach is necessary, along with specific studies in this area. Recently, with advanced technologies, monitoring real-time data has made it easier to enhance the visibility and traceability of the supply chain. Data

mining approaches can be utilized in cold chain management, particularly for detecting outliers. Advanced algorithms are required for analyzing large volumes of data collected from sensors and monitoring devices.

Real-time temperature monitoring and the ability to quickly identify points of disruption in the cold chain are essential to prevent product degradation, ensure regulatory compliance, and improve operational efficiency. While basic statistical methods are common, there is a gap in employing advanced machine learning techniques for cold chain anomaly detection. There is a need for advanced analytical approaches to cope with the growing volume, variety and velocity of cold chain data and growing predictive uncertainty due to unexpected environmental factors. The challenges stem from two major reasons. First, advanced recent technologies including enhanced sensors, IoT integration, wireless communication technologies, and cloud computing lead to a significant big data load. Second, as the operations expand geographically, data scalability of location-based data poses as a significant challenge for outlier analytics. Traditional outlier detection models struggle to address the growing complexities and scale of cold chain data. Advanced adaptive and scalable approaches are essential to ensure the integrity of temperature-sensitive products throughout the supply chain. In order to contribute to this area, a classification-based outlier detection framework, called hybrid Route Detection-based Support Vector Regression (RD-SVR) model, was developed to effectively clean big data and detect outliers based on machine learning. The proposed perspective provides a structured way to tackle the complexities of location-based cold chain data and performs big data outlier detection in a route-based manner with improved accuracy and reliability.

While the RD-SVR model demonstrates high accuracy and robustness in handling temperature deviations in the cold chain, it is necessary to consider its computational efficiency, particularly when handling large data sets. The computational time of the RD-SVR model is higher than other machine learning methods only in the training phase. However, it is more critical to maintain stable and reliable performance for real-time cold chain operations. Besides, since the case study company has high-performance computers, computational time is not a major concern for them.

We believe that this study contributes significantly to both theory and practice. Theoretically, no similar studies using classification-based outlier detection have been encountered in the literature on the cold chain; thus, this study contributes to filling that gap by developing

highly suitable model for the nature of the problem in this field. From a practical standpoint, this study is a case study, demonstrating a real-world application within the company, which makes its implementation both relevant and feasible. The pharmaceutical products transported in the cold chain are highly expensive and include some for rare diseases. Preventing disruptions in the cold chain is crucial for the company's costs. Therefore, this study helps mitigate potential cold chain disruptions, reducing costs for the company and minimizing health risks for the patients.

Although two vehicles from the company provide good sampling for this study, working with data from more vehicles or collaborating with different logistics companies would expand the scope of the research for future works. Beyond its current applications, the RD-SVR model has a high potential to be adopted by different cold chain industries such as food and vaccine distributions. These sectors require sensitive temperature monitoring and predictive capabilities to ensure product safety. However, the food cold chain may require additional concerns such as humidity and vibrations. In the pharmaceutical cold chain, humidity and vibration can be disregarded due to packaging regulatory. The robustness and the adaptive structure of RD-SVR makes it particularly suitable for handling the challenges of various industries, including fluctuating environmental conditions and missing sensor data.

CRedit authorship contribution statement

Sevde Ceren Yıldız Özenc: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Merve Er:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization. **Seniye Ümit Fırat:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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